



CANDIDATE
NAME

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CIVICS
GROUP

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REGISTRATION
NUMBER

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H2 Biology

9744/02

Paper 2 Structured Questions

14 September 2018

2 hours

Candidates are to **answer all questions** in this question booklet.

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
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Total	100

This document consists of **29** printed pages and **1** blank page..

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Section A

Answer **all** the questions in this section.

- 1 The synthesis of collagen is shown in Fig. 1.1.

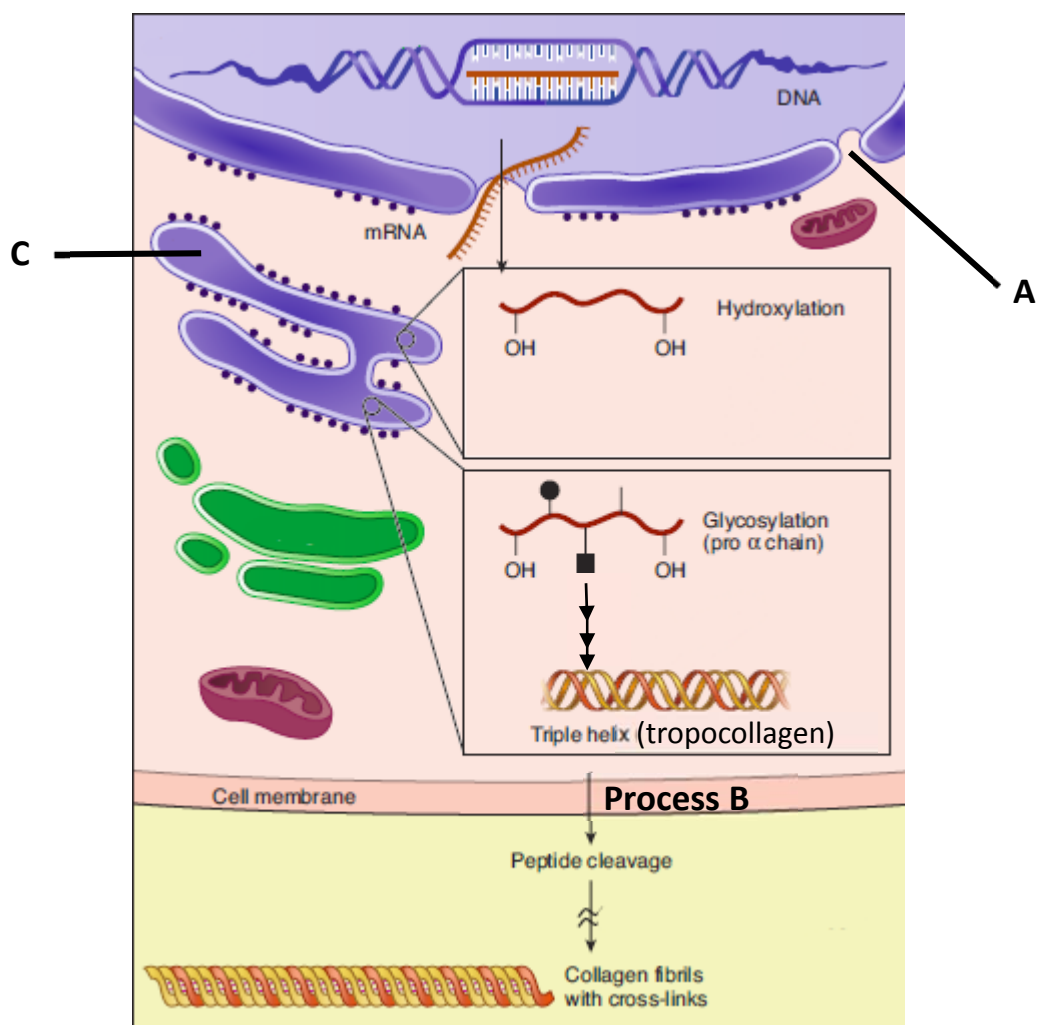


Fig. 1.1

- (a) (i) Describe two important functions of structure A in the synthesis of collagen.

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.....[2]

- (ii) Tropocollagen leaves the cell to be assembled to form collagen fibrils via **Process B**. Outline **Process B**.

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- (b) Suggest how chemical modification such as hydroxylation in organelle **C** results in collagen having a high tensile strength.

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Fig. 1.2 shows two electron micrographs. One of the electron micrograph shows part of a normal cell while the other electron micrograph shows part of a cancer cell. The white arrows point to an organelle within the cell. The appearance of this organelle in both cell types were visibly different. The cancer cell had a higher activity than the normal cell.

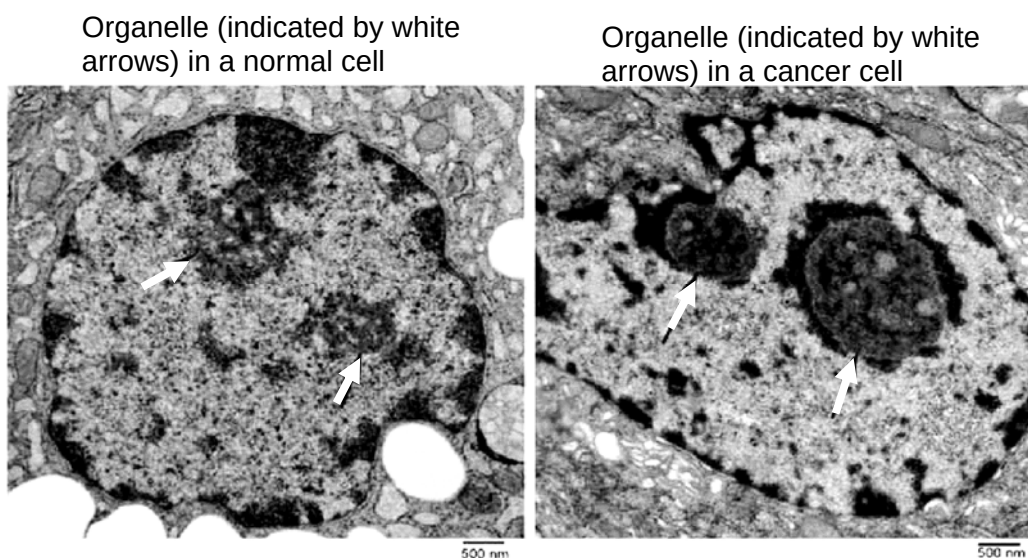


Fig. 1.2

- (c) (i) Describe the visible difference between the organelle indicated by the white arrows in Fig. 1.2, between the cancer cell and the normal cell.

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.....[1]

- (ii) Account for the higher activity of the cancer cell.

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[Total: 10]

- 2 The rate of glycolysis is regulated by the action of ATP on the enzyme phosphofructokinase (PFK), the third enzyme in the glycolysis pathway. PFK catalyzes the formation of fructose 1,6-bisphosphate from fructose-6-phosphate and ATP.

A graph of PFK against F6P concentration would exhibit the 'sigmoidal curve' typical of allosteric enzymes.

Fig. 2.1 shows the effect of substrate F6P (fructose 6-phosphate) concentration on the activity of PFK at different ATP concentrations. PFK is an allosteric enzyme with a quaternary structure that is inhibited by high levels of ATP. High levels of ADP and AMP will increase activity of the enzyme.

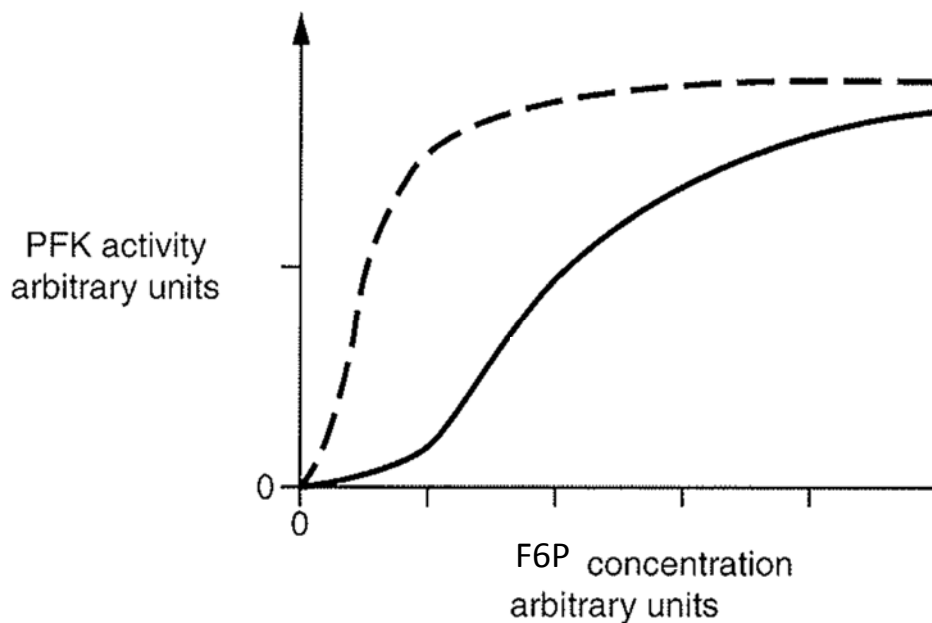


Fig. 2.1

- (a) (i) With reference to PFK, explain the term '*allosteric enzyme*'.
-
-[1]
- (ii) On Fig. 2.1, label the graphs which reflect low ATP and high ATP concentrations respectively. [1]
- (b) Enzyme researchers have used a model of allosteric enzyme mechanism called the 'symmetry model' to explain the action of PFK. F6P binds with great affinity only to the active state but not to the inactive state. Binding of one F6P molecule will progressively shift PFK structure from the inactive state to the active state.

A graph of PFK against F6P concentration would exhibit the 'sigmoidal curve' typical of allosteric enzymes.

- (i) Explain how the binding of a molecule like AMP to PFK can increase the activity of PFK.

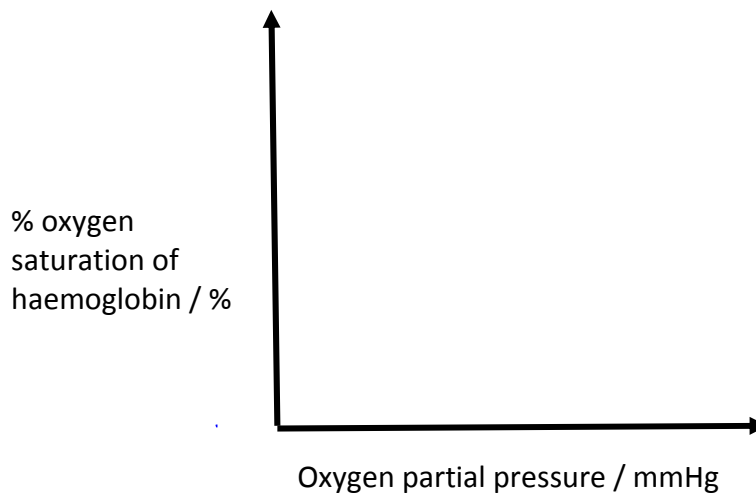
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- (ii) The enzymatic mechanism in PFK is similar to that of the oxygen-binding mechanism of the transport protein, haemoglobin. Sketch the shape of the oxygen binding graph for haemoglobin at increasing concentrations of oxygen in the space below. [1]



- (iii) Using your knowledge of the structure of haemoglobin, explain the shape of the graph you drew in (b) (ii).

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- (c) Explain the significance of regulation of PFK by ATP to AMP ratios.

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[Total: 10]

- 3 Bone marrow contains stem cells that divide by mitosis to form blood cells. The fate of a stem cell was tracked and it was recorded that during the observed duration the stem cell divided asymmetrically each time.

Fig. 3.1 shows changes in the mass of DNA in a human stem cell from bone marrow during three cell cycles.

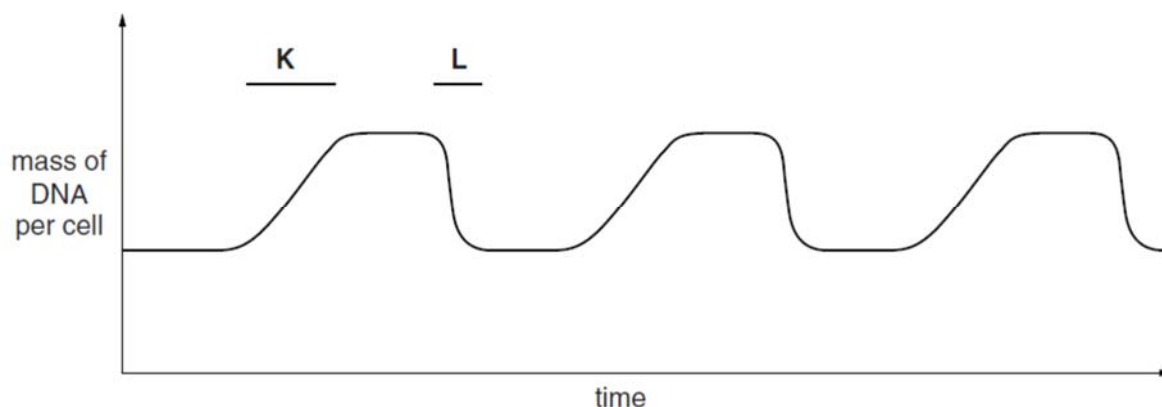


Fig. 3.1

- (a) With reference to the information provided above,
- Describe what happens to bring about the changes in the mass of DNA per cell at time period **K** and at time period **L**.
K

L
[2]
 - State one function of these stem cells undergoing the above type of cell division.

[1]
 - The process of meiosis is significant to natural selection in evolution.
 Explain this significance.

[2]

A bone marrow cell was extracted and observed under the electron micrograph shown in Fig. 3.2. The student focused on an organelle which he described as having “an envelope surrounding genetic material containing both darker and lighter stained patches, distinct from the site where ribosomal subunits were assembled”.

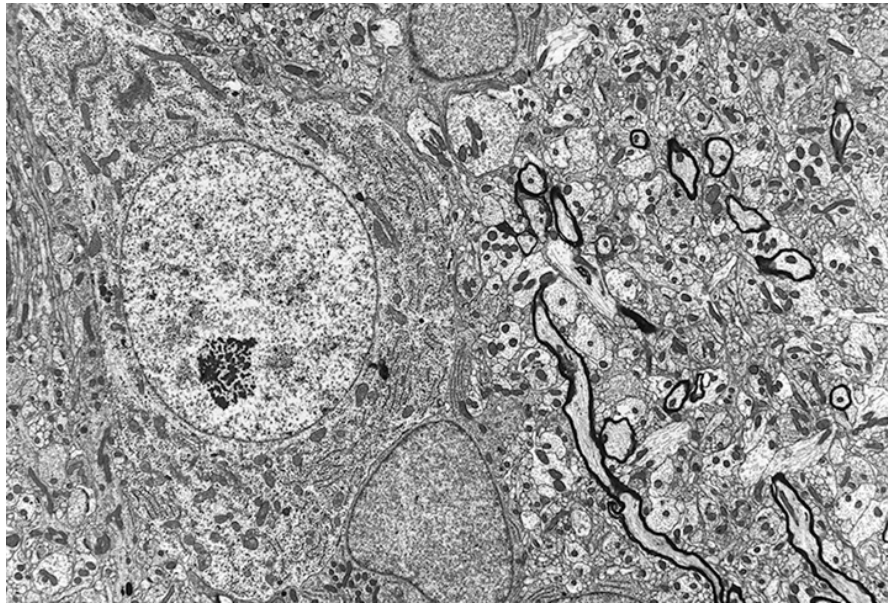


Fig. 3.2

- (b)** Explain the significance of the “darker and lighter stained patches” that the student referred to, in a cell undergoing differentiation.

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.....[3]

The use of embryonic stem cells (ESCs) for stem cell therapy and research is controversial and considered by many people as unethical. Scientists have circumvented this issue through the use of induced pluripotent stem cells (iPSCs) as an alternative to ESCs.

Fig. 3.3 summarises the procedure for obtaining iPSCs and its use.

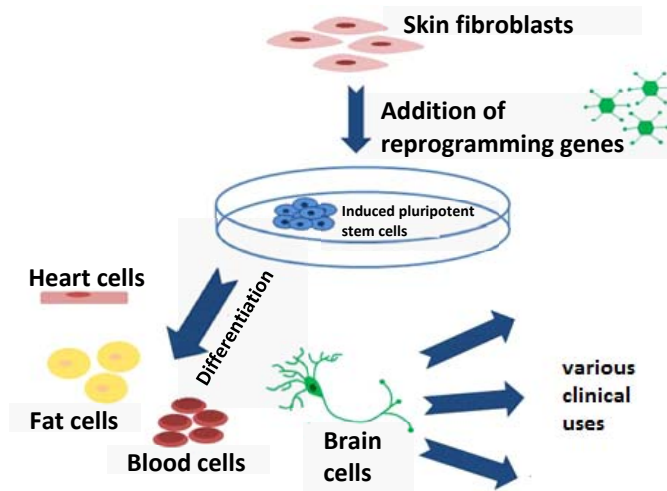


Fig. 3.3

(c) Explain why the use of iPSCs is preferred over ESCs.

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[Total: 10]

- 4 Fig. 4.1 shows part of a DNA molecule.

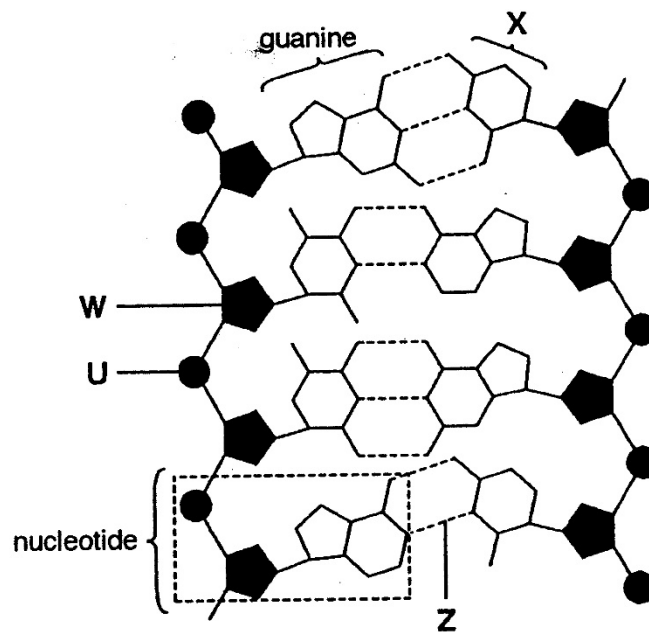


Fig. 4.1

- (a) (i) Name **W** to **Z**. [2]

W

U

X

Z

- (ii) Give **one** advantage of DNA having two strands.

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[1]

- (b) Fig. 4.2 shows a linear chromosome undergoing the first round of DNA replication.

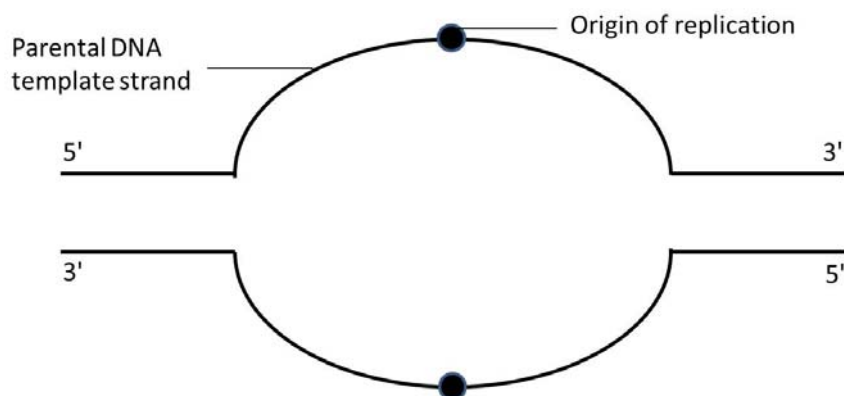


Fig. 4.2

- (i) On Fig. 4.2, draw the direction of DNA synthesis for the leading strands using solid arrows ($\rightarrow$) and for the lagging strands using dashed arrows ($- - - >$). [1]
- (ii) State **two** ways DNA replication is different from transcription.

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- (c) Fig. 4.3 is an electronmicrograph showing the transcription of genes for ribosomal RNAs at adjacent positions along the chromosome in a eukaryotic cell.

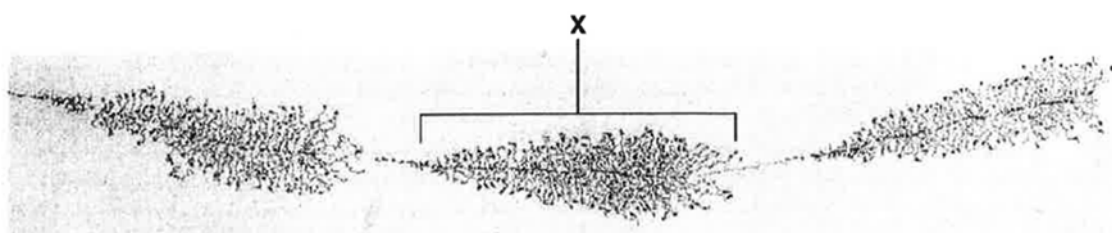


Fig. 4.3

- (i) Describe and explain the pattern of transcription visible on the part of the DNA coding for ribosomal RNA, labelled X in Fig. 4.3.

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- (ii) Suggest possible roles for the non-transcribed DNA that is found between the ribosomal genes.

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[Total:10]

- 5 Human immunodeficiency virus (HIV) causes the condition known as acquired immunodeficiency syndrome (AIDS) in humans.

The development of AIDS in HIV infected persons may take place many years after infection with the virus, e.g. 8.5 years later, as illustrated in Fig. 5.1.

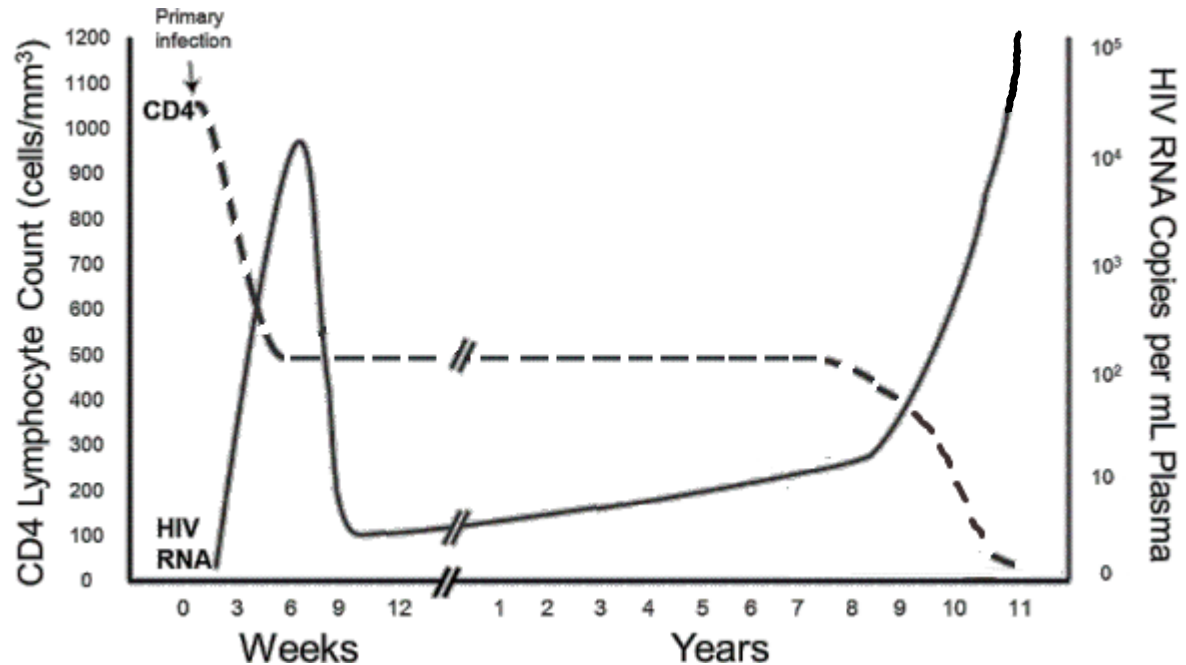


Fig. 5.1

- (a) With reference to Fig. 5.1,
- (i) Explain the level of HIV RNA detected in the plasma from week 9 onwards after infection.

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- (ii) Explain why AIDS patients have compromised immune systems.

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- (iii) With a weakened immune system, AIDS patients are susceptible to opportunistic infections, whereby infections occur more frequently and are more severe.

On Fig. 5.1, indicate clearly with an arrow along the X axis when HIV infected patients succumb to opportunistic infections. [1]

- (b) Scientists are trying to develop an effective vaccine to protect people against HIV. However, there are three main problems: HIV causes immune system of infected individuals to be compromised, HIV rapidly enters host cells and HIV has a high mutation rate.

- (i) State one structural feature of HIV that accounts for its high mutation rate and explain why this means that a vaccine might not be effective against HIV.

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- (ii) State the type of immunity conferred by vaccines.

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[Total: 10]

- 6** A plant breeder crossed a plant from a pure-bred line of tomato plants with red uniformly pigmented fruit with a plant from a pure-bred line of plants with orange-coloured fruit, but with unattractive dark patches at the base of the unripe fruit. The resulting generation all produced red fruit with dark patches. Plants from this generation were interbred. The resulting progeny showed the following numbers of plants in each of the three phenotypes:

Red fruit with dark patches	98
Red uniformly pigmented fruit	46
Orange fruit with dark patches	44

- (a)** Explain how genotype is linked to the phenotype.

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- (b)** The genes involved in the cross above are completely linked. Explain the meaning of the term 'complete linkage'.

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- (c) Draw a genetic diagram to show how the second cross could lead to the three phenotypes.

[5]

The χ^2 distribution table (Table 6.1) and equation to calculate χ^2 is shown below. Using the formula, the calculated χ^2 value for the cross was 0.38.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Table 6.1

Degree of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (d)** Using the calculated value of χ^2 and the table of probabilities provided in Table 6.1, explain what conclusion can be drawn from the data.

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[Total: 11]

- 7 Fig. 7.1 shows a CREB (cAMP response element binding) signalling pathway. This pathway is triggered when ligand X binds to a receptor at the cell membrane.

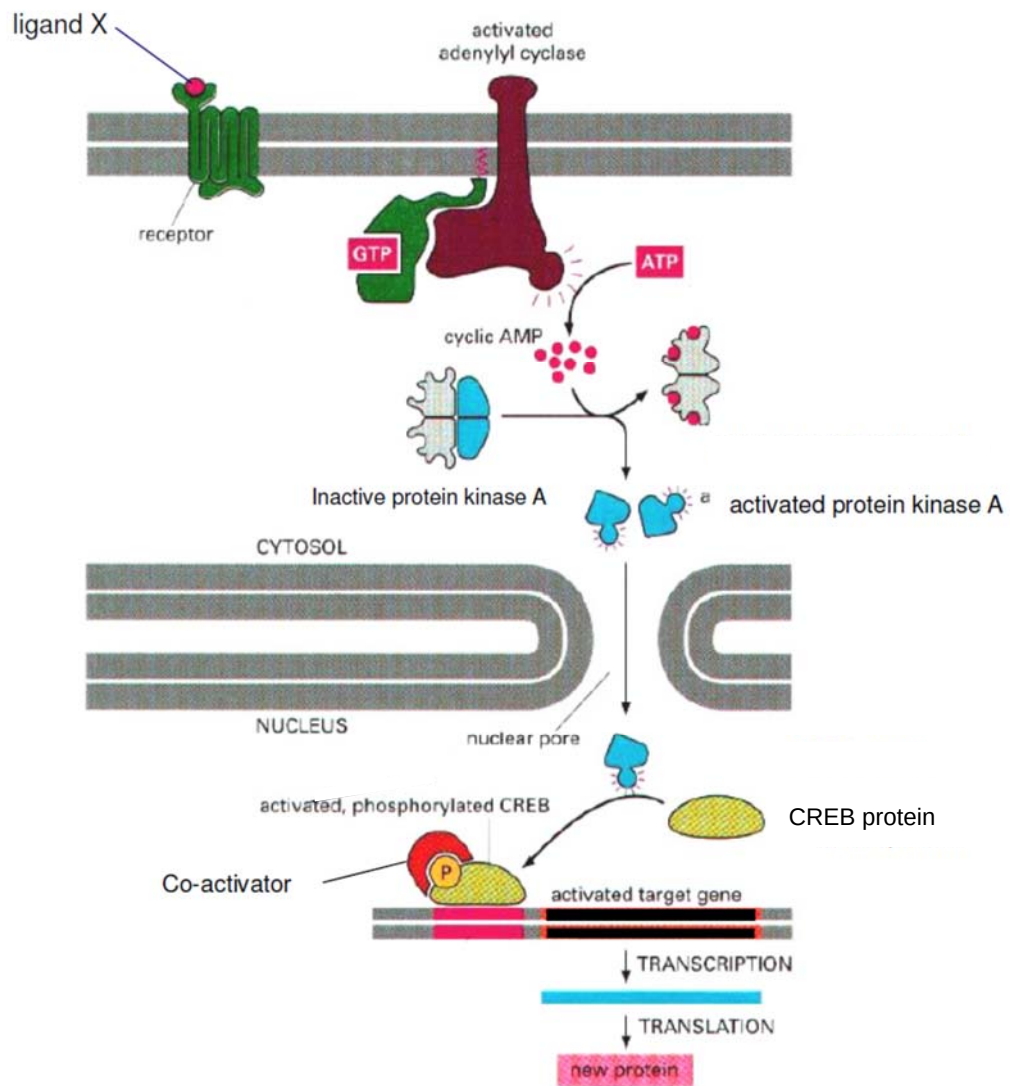


Fig. 7.1

- (a) With reference to Fig. 7.1,
- (i) Describe precisely the structure of the G protein coupled receptor and explain how the described structure enables it to perform its function.

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- (ii) Describe how protein kinase A (PKA) is activated.

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- (iii) State the role of PKA in this signaling pathway?

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- (b) CREB is an activator protein that binds to its specific control element to upregulate the transcription of the target gene.

Explain how an activator protein like CREB can upregulate transcription of a gene.

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- (c) CREB is known to be involved in other signaling pathways. With reference to Fig. 7.1 or otherwise, suggest how it is possible for CREB to bind to other control elements so as to regulate the transcription of different genes.

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[Total: 9]

- 8 A student investigated the effect of different wavelengths of light on the rate of photosynthesis. She used the apparatus shown in Fig. 8.1 below.

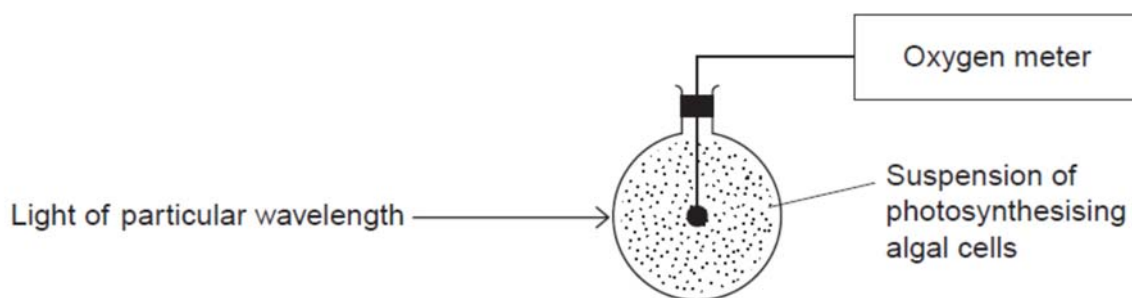


Fig. 8.1

- (a) State a condition that is to be kept optimum in the investigation shown in Fig. 8.1.
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-[1]
- (b) The student did not use a buffer to maintain the pH of the solution and observed that the pH of the solution changed over the course of the investigation.

Predict and explain the effect of this pH change on photosynthesis in the investigation.

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.....[2]

The student plotted her results below in Fig. 8.2.

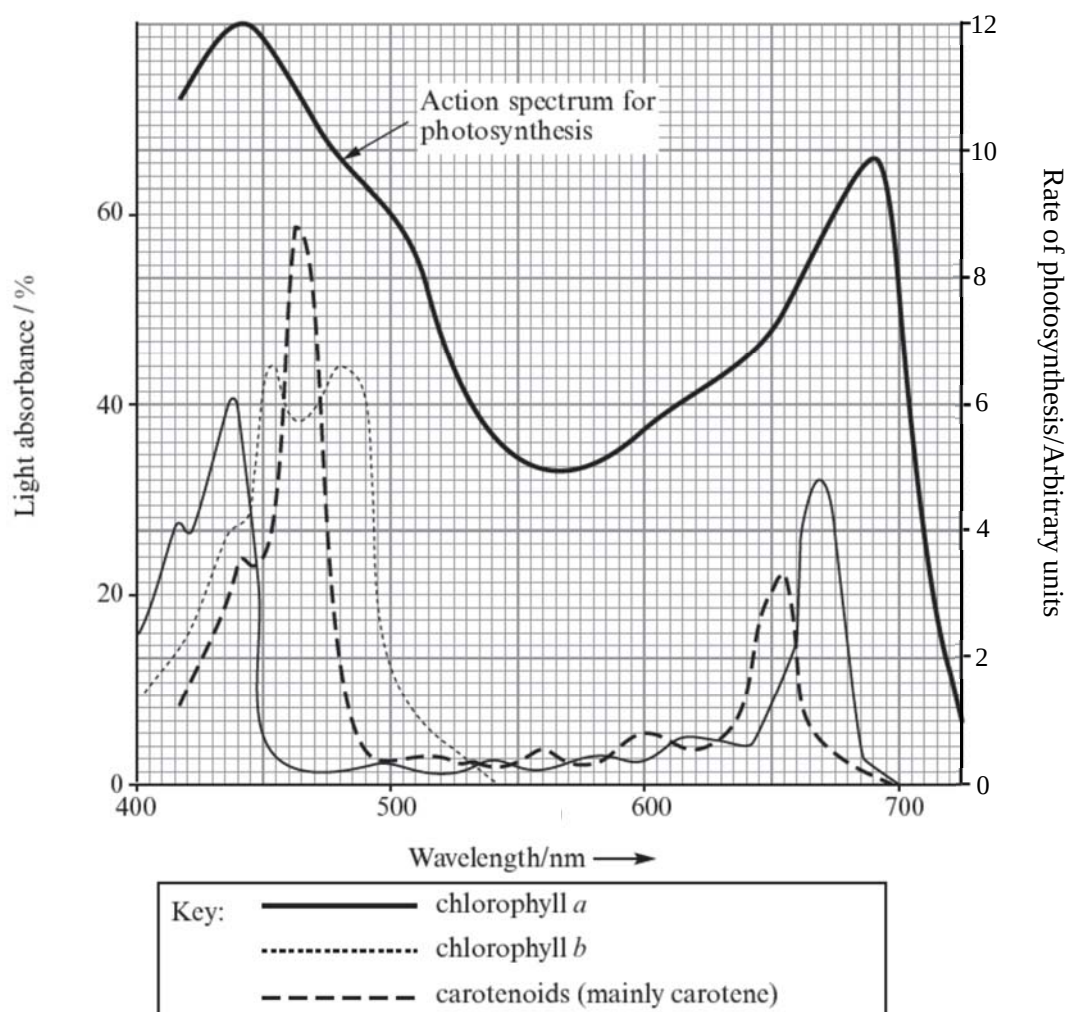


Fig. 8.2

- (c) Chlorophyll a functions as a primary photosynthetic pigment. With reference to Fig. 8.2, explain the advantage of a plant having accessory pigments like carotenoids.

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- (d) Explain the relationship between the absorption spectrum and the action spectrum.

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One of the most significant consequences of climate change is global warming. Fig. 8.3 shows the impact of global warming on crop yield on two temperate cereal crops, corn and soybean in Africa.

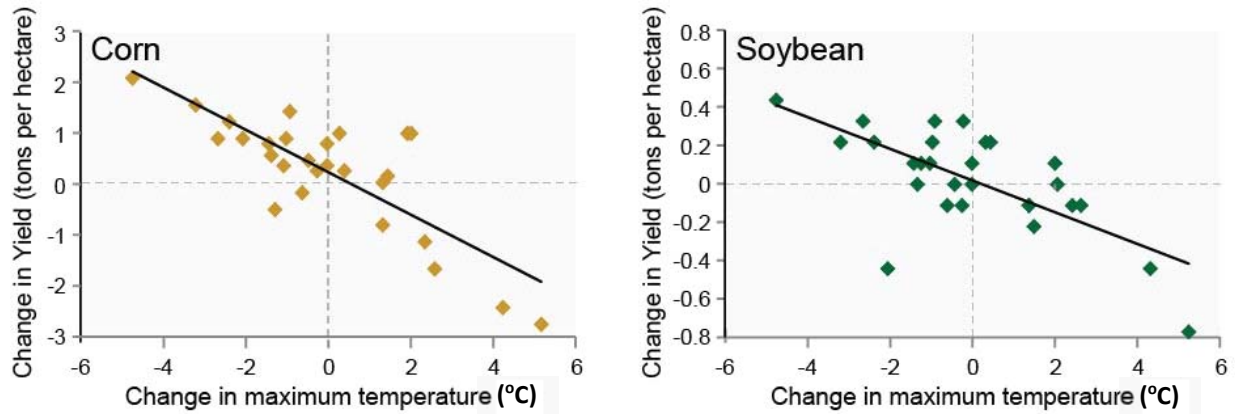


Fig. 8.3

- (e) Describe and explain how increase in temperatures brought about by global warming impacts the crop yield of corn and soybean.

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[Total: 10]

- 9 A student carried out an investigation on yeast respiration using the following setup shown in Fig. 9.1. She put 5 grams of yeast into a glucose solution and put a layer of oil on top of the mixture. This layer of oil is impermeable to oxygen but not to carbon dioxide. The total volume of gas was recorded every 10 minutes for 1 hour and the results are shown in Table 9.1.

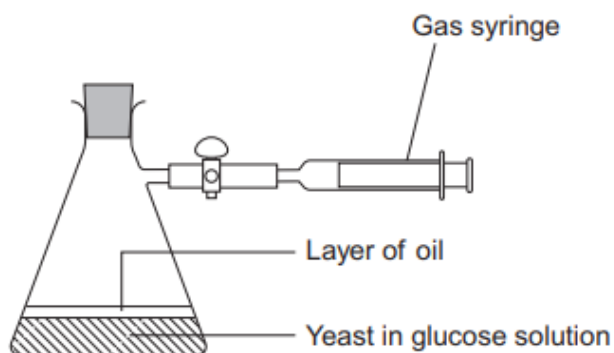


Fig. 9.1

Table 9.1

Time / minutes	Total volume of gas collected / cm ³
10	0.3
20	0.9
30	1.9
40	3.1
50	5.0
60	5.2

- (a) ii) Calculate the rate of gas production in cm³ g⁻¹ min⁻¹ during the first 40 minutes of this investigation. Show your working clearly. [1]

Answer = cm³ g⁻¹ min⁻¹

- ii) Suggest why the rate of gas production decreased between 50 and 60 minutes.

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[1]

- (b) The Respiratory Quotient (RQ) is defined as the ratio of carbon dioxide produced to oxygen consumed per unit time by an organism and the formula is as follows:

$$RQ = \frac{\text{volume of CO}_2 \text{ produced}}{\text{volume of O}_2 \text{ consumed}} \text{ per unit time}$$

Given that the rate of carbon dioxide produced equals to the rate of oxygen used for aerobic respiration, suggest and explain the RQ value for the setup shown in Fig. 9.1.

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[2]

- (c) To show that the presence of oil has an effect on yeast respiration, the student also created another similar setup as the one shown in Fig. 9.1. All variables were kept constant in this second setup except that no oil was used.

Explain if there is any difference in the amount of ATP produced between both setups.

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[3]

In another experiment, the student investigated the effect of temperature on the rate of oxygen consumption of the lizard, *Sauromalus hispidus*. The body temperature of a lizard varies with environmental temperature.

Several baby lizards were fitted with small, airtight masks that covered their heads. Air was supplied inside the mask through one tube, and collected through another. The differences between oxygen concentrations in the air supplied for inhalation and the exhaled air enabled the researchers to measure the rate of oxygen consumption of the lizards.

The rate of oxygen consumption of each baby lizard was measured at different temperatures ranging from 15 °C to 40 °C and the average measurements were computed. The results shown in Fig. 9.2.

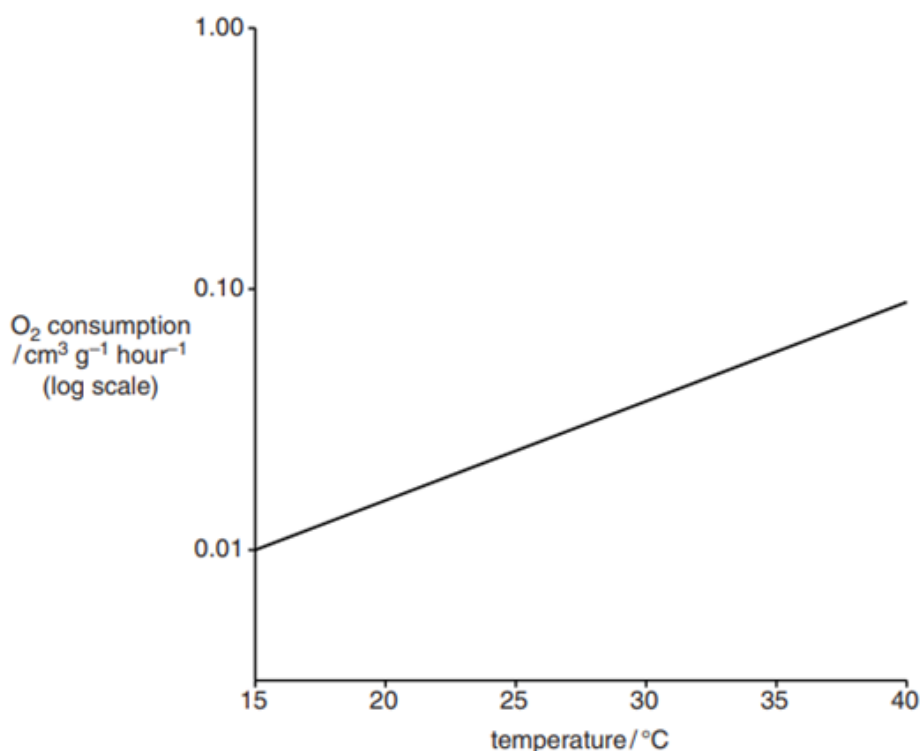


Fig. 9.2

- (d) Using Fig. 9.2, suggest and explain how temperature influences the development of baby lizards.

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[Total: 10]

- 10** The theory of continental drift was proposed over 90 years ago by Alfred Wegener and is now a widely accepted theory of how the continents were formed.

The supercontinent Gondwana is shown in Fig. 10.1. The land that was to form New Zealand began to break away from the supercontinent, Gondwana around 80 million years ago and broke away totally around 65 million years ago (Fig. 10.2). Ever since, New Zealand is now composed of North Island and South Island (Fig. 10.3).

The separation of New Zealand from Gondwana has led to some species like the New Zealand kiwi bird being distantly related to the rhea (South America), ostrich (Africa) and cassowary (Australia).



Fig. 10.1



Fig. 10.2

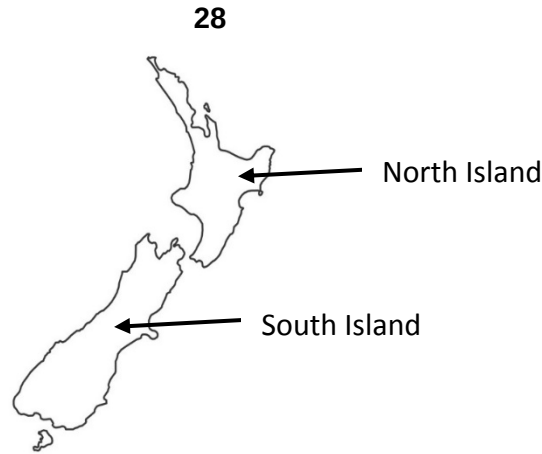


Fig. 10.3

- (a) Using your knowledge of biogeography and continental drift, outline how the kiwi could be distantly related to some birds from other continents.

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For reasons that are not apparent, New Zealand was not inhabited by many mammals. Instead, its fauna became dominated by birds and insects. Birds became the predators, the scavengers, the herbivores and the insectivores. They lived everywhere from the highest mountains to the sea. Some have lost the ability to fly and became ground dwellers.

The kakapo, kea and kaka are parrots that are currently found in New Zealand (Fig. 10.4).

The kakapo (*Stringops habroptila*) is found only in New Zealand and is flightless. Highly endangered, they were relocated to several islands in New Zealand during the 1980s and 1990s.

The kea (*Nestor notabilis*) is the world's only alpine parrot, it can be found living near forested mountain habitats - Southern Alps on South Island. Kea are not found on North Island now.

Kaka (*Nestor meridionalis*) can be found on South Island and the sub-species of Kaka (*Nestor meridionalis septentrionalis*) is found on North Island. All Kaka live in the lowlands and mid-altitude regions.



kakapo



kea



kaka

Fig. 10.4

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-
- [1]

Mya = Million years ago

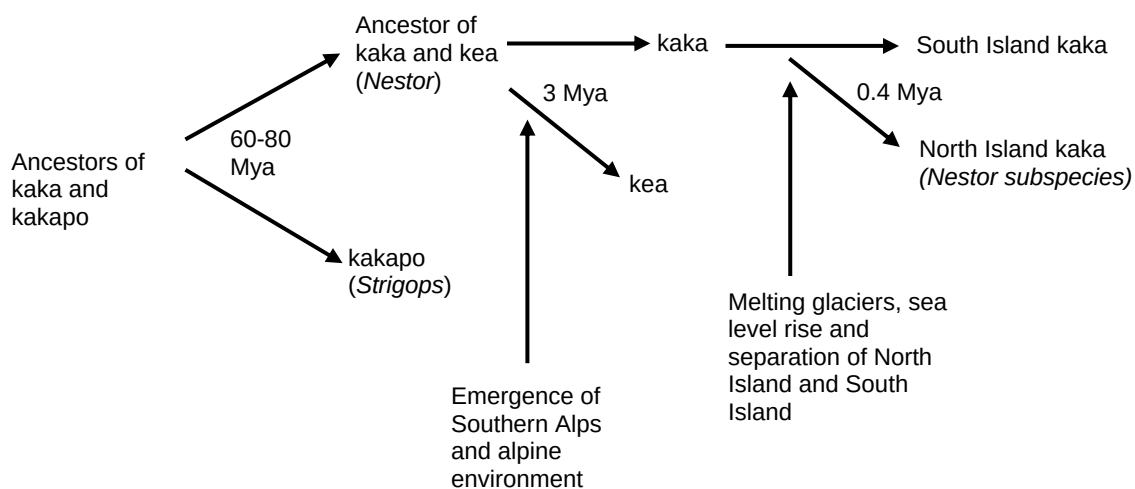


Fig. 10.5

-[5]

- (d) With reference to Fig. 10.5, suggest a possible reason for the emergence of a sub-species of kaka on North Island.

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[Total: 10]

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